IDS 702 HW 3

KEY

Instructions: Use this template to complete your assignment. When you click "Render," you should get a PDF document that contains both your answers and code. You must show your work/justify your answers to receive credit. Submit your rendered PDF file on Gradescope. **Remember to render frequently**, as this will help you to catch errors in your code before the last minute. You must show your work for all problems, and you must provide a written answer for all problems. For example, you should write "There are XX observations and each observation represents a ________"; it is insufficient to just show the code.

Add your name in the Author section in the header

Load Data

```
library(tidyverse)
library(gridExtra)

colleges <- read.csv("https://raw.githubusercontent.com/anlane611/datasets/refs/heads/main/c
```

Part 1

1

a. 797 observations and each observation is a college/university. (No code necessarily required, and can be something like `glimpse()` instead of `nrow()`)

```
nrow(colleges)
```

```
[1] 797
```

b.

```r
summary(colleges[,c("control","basic","student_count","ft_pct",
                    "med_sat_value","endow_value",
                    "grad_100_value","grad_150_value",
                    "retain_value")])
```

```
   control             basic           student_count       ft_pct
 Length:797         Length:797         Min.   :   82   Min.   :  6.00
 Class :character   Class :character   1st Qu.:  979   1st Qu.: 80.50
 Mode  :character   Mode  :character   Median : 1677   Median : 91.90
                                       Mean   : 5544   Mean   : 86.41
                                       3rd Qu.: 5156   3rd Qu.: 97.20
                                       Max.   :51333   Max.   :100.00

 med_sat_value   endow_value      grad_100_value grad_150_value
 Min.   : 666   Min.   :     28   Min.   : 0.00   Min.   : 0.00
 1st Qu.: 990   1st Qu.:   7964   1st Qu.:23.60   1st Qu.:40.52
 Median :1081   Median :  20869   Median :38.00   Median :55.55
 Mean   :1099   Mean   :  78374   Mean   :41.97   Mean   :55.87
 3rd Qu.:1192   3rd Qu.:  55975   3rd Qu.:59.55   3rd Qu.:72.15
 Max.   :1534   Max.   :2505435   Max.   :92.80   Max.   :97.80
 NA's   :168    NA's   :71        NA's   :7       NA's   :7
  retain_value
 Min.   :  0.00
 1st Qu.: 65.80
 Median : 76.10
 Mean   : 74.48
 3rd Qu.: 86.60
 Max.   :100.00
 NA's   :5
```

The following variables have missing values: med_sat_value (168), endow_value (71), grad_100_value (7), grad_150_value (7), and retain_value (5).

```r
colleges_full <- colleges |>
  filter(complete.cases(control, basic, student_count,
                        ft_pct, med_sat_value, endow_value,
                        grad_100_value, grad_150_value,
                        retain_value))
```

2

```
colleges_full |>
  count(basic) |>
  mutate(prop=n/sum(n))
```

```
                                          basic   n      prop
1            Baccalaureate Colleges--Arts & Sciences 195 0.3170732
2             Baccalaureate Colleges--Diverse Fields 231 0.3756098
3      Research Universities--high research activity  84 0.1365854
4 Research Universities--very high research activity 105 0.1707317
```

```
colleges_full |>
  count(control) |>
  mutate(prop=n/sum(n))
```

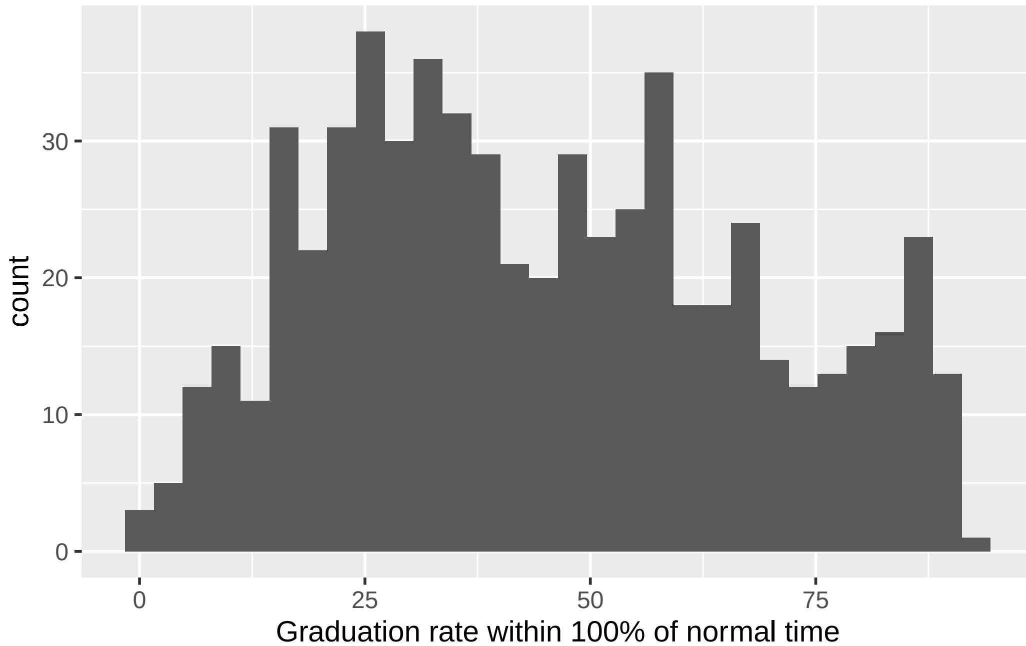
```
                control   n      prop
1 Private not-for-profit 409 0.6650407
2                 Public 206 0.3349593
```

	Count (n)	Proportion or %
Basic classification		
Baccalaureate Colleges–Arts & Sciences	195	32
Baccalaureate Colleges–Diverse Fields	231	38
Research Universities–high research activity	84	14
Research Universities–very high research activity	105	17
Control of institution		
Private not-for-profit	409	67
Public	206	33

3

a.

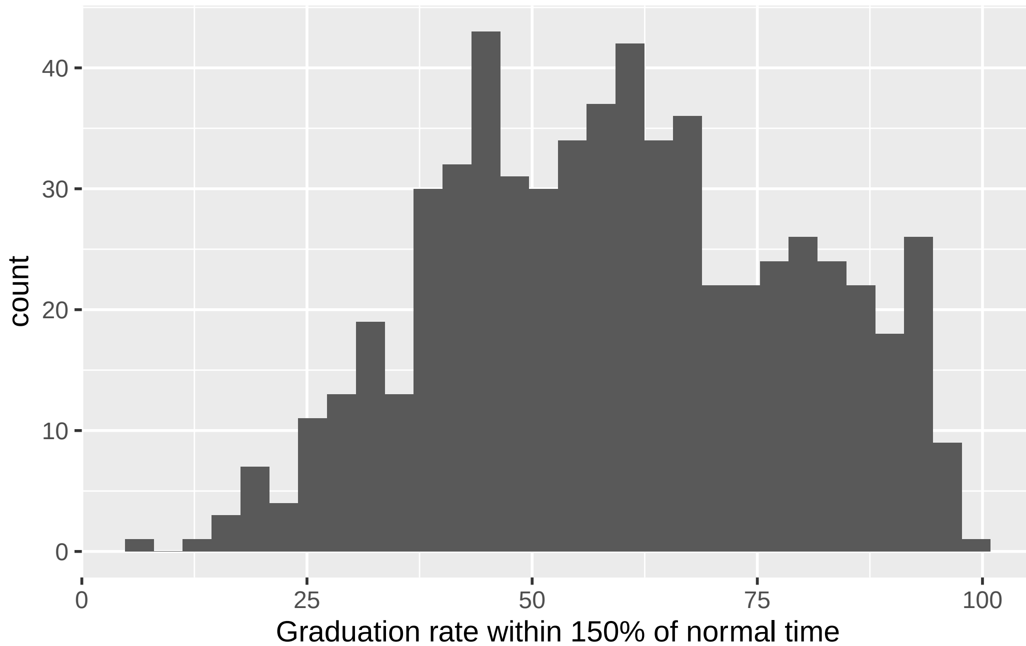
```
ggplot(colleges_full, aes(x=grad_100_value))+
  geom_histogram()+
  labs(x="Graduation rate within 100% of normal time")
```



The distribution is relatively evenly spread across the full range 0-100, which is not necessarily what we would expect to see. Many institutions have graduation rates below 50%, which is lower than we might expect.

b.

```
ggplot(colleges_full, aes(x=grad_150_value))+
  geom_histogram()+
  labs(x="Graduation rate within 150% of normal time")
```




```r
colleges_full[which(colleges_full$chronname=="Northeastern University"),
              c("grad_100_value","grad_150_value")]
```

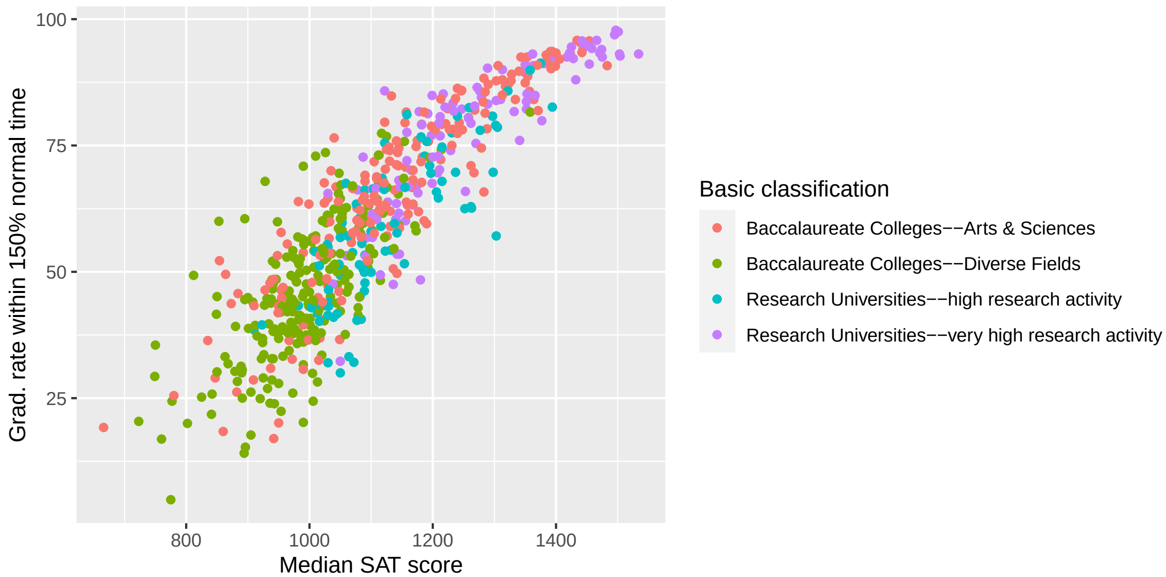
```
    grad_100_value grad_150_value
237              0           82.6
```

The distribution of graduation rate within 150% of normal time looks more like what we would expect, with a large proportion of institutions having a rate above ~40%. Looking at Northwestern, we see that the grad_100_value is 0, while grad_150_value is 82.6. This is likely a data error, so grad_150_value seems to have more reliable data.

4

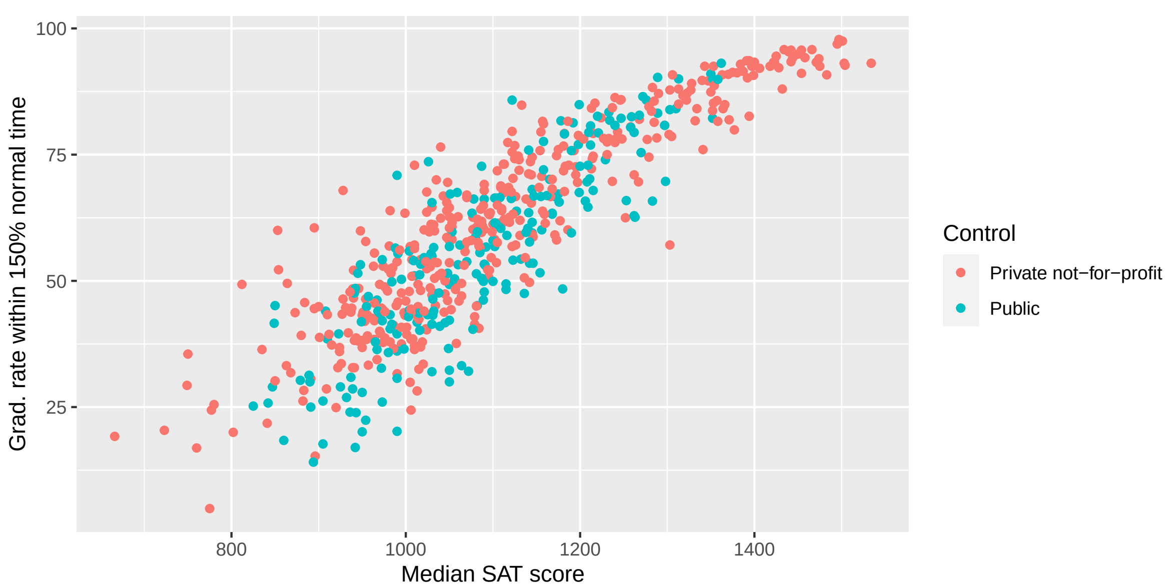
```r
colleges_full$basic_fac <- factor(colleges_full$basic)
colleges_full$control_fac <- factor(colleges_full$control)

ggplot(colleges_full, aes(x=med_sat_value, y=grad_150_value,
                                   col=basic_fac))+
  geom_point()+
  labs(x="Median SAT score",y="Grad. rate within 150% normal time",
       col="Basic classification")
```



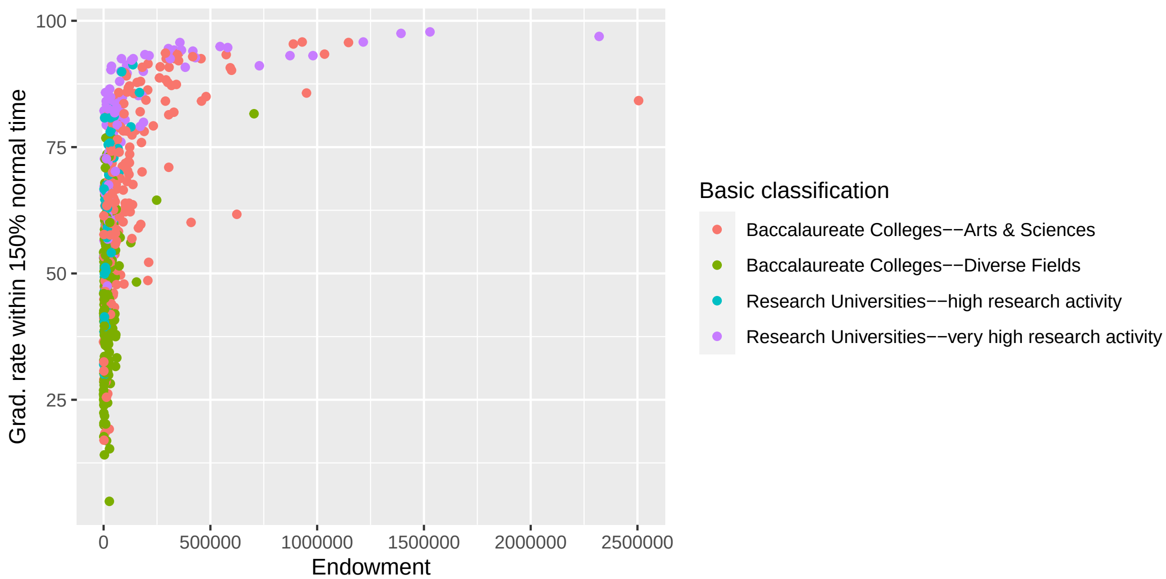
Graduation rates increase as median SAT score increases, generally, and the research universities tend to have higher median SAT scores and graduation rates.

```
ggplot(colleges_full, aes(x=med_sat_value, y=grad_150_value,
                                    col=control_fac))+
  geom_point()+
  labs(x="Median SAT score",y="Grad. rate within 150% normal time",
       col="Control")
```



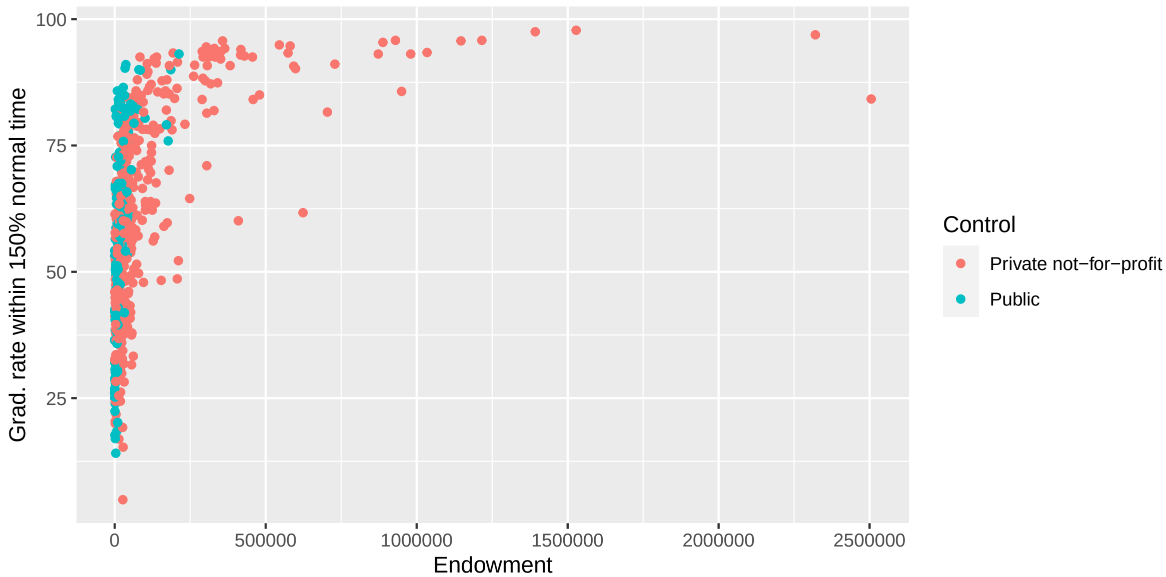
Graduation rates increase as median SAT score increases, generally, and there doesn't seem to be a difference between public and private schools.

```
ggplot(colleges_full, aes(x=endow_value, y=grad_150_value,
                                   col=basic_fac))+
  geom_point()+
  labs(x="Endowment",y="Grad. rate within 150% normal time",
       col="Basic classification")
```



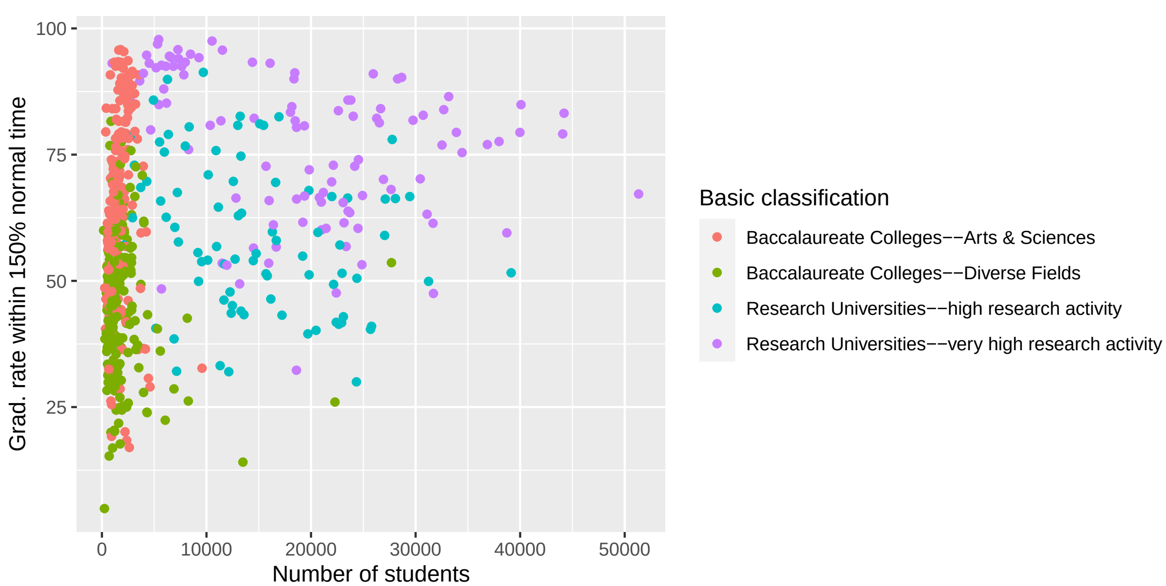
There is not a clear trend because many schools have a small endowment and a few have large endowments.

```
ggplot(colleges_full, aes(x=endow_value, y=grad_150_value,
                                   col=control_fac))+
  geom_point()+
  labs(x="Endowment",y="Grad. rate within 150% normal time",
       col="Control")
```



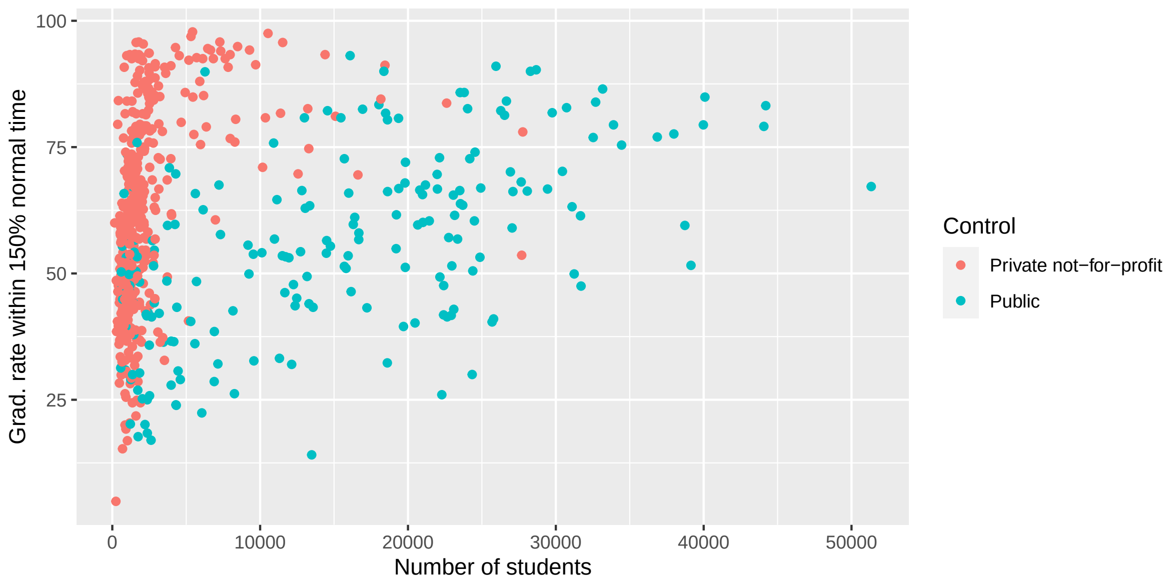
The schools with large endowments are private, and the schools with large endowments also have high graduation rates.

```
ggplot(colleges_full, aes(x=student_count, y=grad_150_value,
                                             col=basic_fac))+
  geom_point()+
  labs(x="Number of students",y="Grad. rate within 150% normal time",
       col="Basic classification")
```



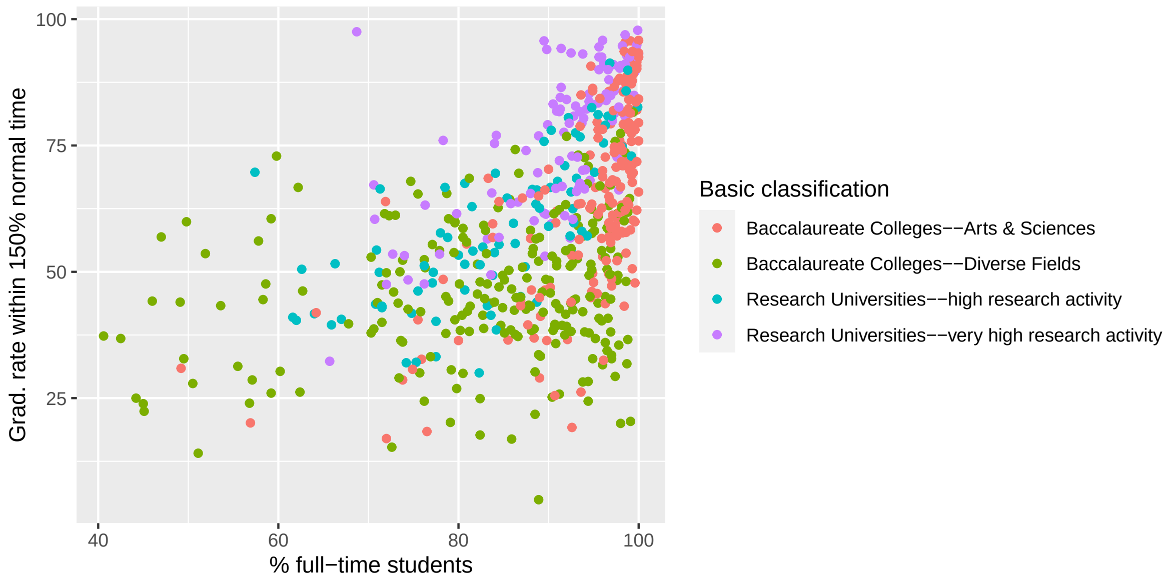
The research universities have higher enrollments, and there doesn't seem to be a clear trend for enrollment count and graduation rates for public vs private schools.

```
ggplot(colleges_full, aes(x=student_count, y=grad_150_value,
                                        col=control_fac))+
  geom_point()+
  labs(x="Number of students",y="Grad. rate within 150% normal time",
       col="Control")
```



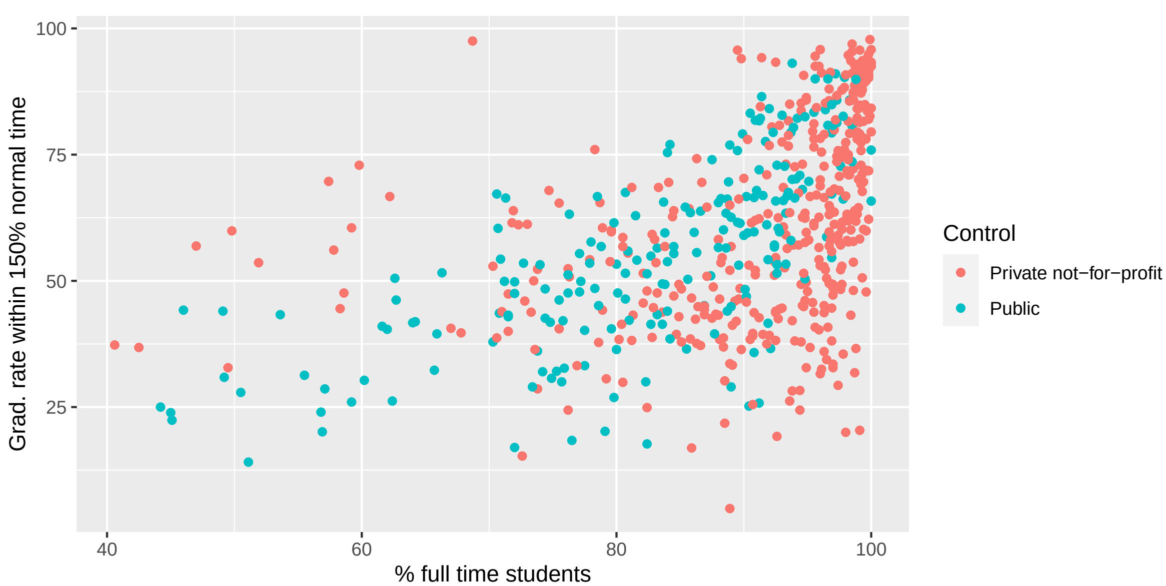
The public schools have higher enrollment numbers, and there doesn't seem to be a clear trend for enrollment count and graduation rates for public vs private schools.

```
ggplot(colleges_full, aes(x=ft_pct, y=grad_150_value,
                                        col=basic_fac))+
  geom_point()+
  labs(x="% full-time students",y="Grad. rate within 150% normal time",
       col="Basic classification")
```



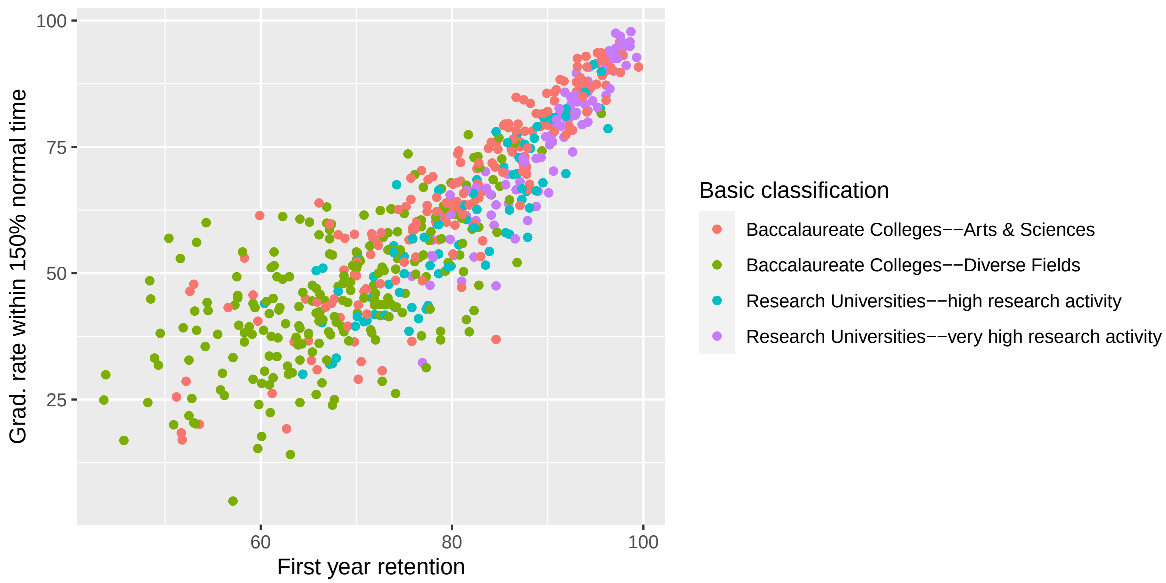
In general, the schools with higher % full-time students have higher graduation rates, though the trend with classification is unclear.

```
ggplot(colleges_full, aes(x=ft_pct, y=grad_150_value,
                                   col=control_fac))+
  geom_point()+
  labs(x="% full time students",y="Grad. rate within 150% normal time",
      col="Control")
```



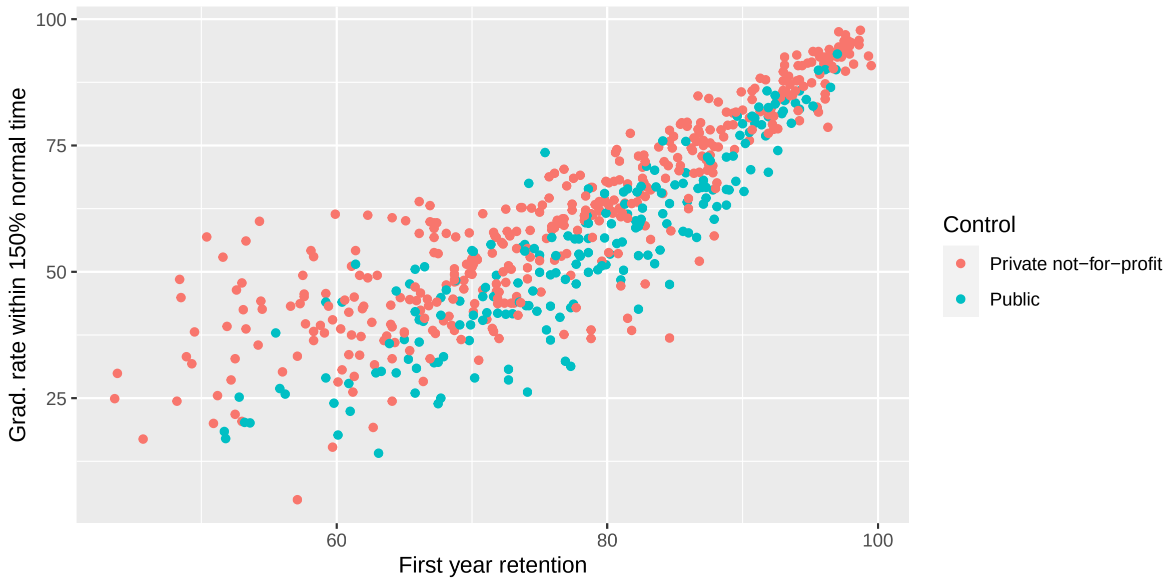
There doesn't seem to be a difference in the relationship between % full-time students and graduation rate for private and public schools.

```
ggplot(colleges_full, aes(x=retain_value, y=grad_150_value,
                                   col=basic_fac))+
  geom_point()+
  labs(x="First year retention",y="Grad. rate within 150% normal time",
       col="Basic classification")
```



Retention and graduation rate seem to have a strong positive relationship, and the research universities have higher retention and graduation rates.

```
ggplot(colleges_full, aes(x=retain_value, y=grad_150_value,
                                   col=control_fac))+
  geom_point()+
  labs(x="First year retention",y="Grad. rate within 150% normal time",
       col="Control")
```



There doesn't seem to be a difference in the relationship between retention and graduation rate for private and public schools.

5

Students can choose either to combine the levels to make interpretation easier, or say that it is better to leave the levels as is because there is sufficient sample size in each level.

6

$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$, $\epsilon \sim N(0, \sigma^2)$

where $Y =$ graduation rate within 150% of normal time

$x_1 =$ student count

$x_2 = 1$ if control=Public, 0 otherwise

7

$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_1 x_2 + \epsilon$, $\epsilon \sim N(0, \sigma^2)$

where $Y =$ graduation rate within 150% of normal time

$x_1 =$ full-time %

$x_2 = 1$ if control=Public, 0 otherwise

$x_3 =$ retention rate

$x_4 =$ median SAT score